An Analysis of Human GLP1R Genetic Variation and Its Potential Impact on Drug Binding

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Main Databases, Tools, and Datasets Used:

Ensembl (GLPR1 Gene)
Gene ID: ENSG00000112164
https://www.ensembl.org/Homo_sapiens/Gene/Summary?db=core;g=ENSG00000112164;r=6:39048781-39091303;t=ENT00000373256

Ensembl Variant Table
https://www.ensembl.org/Homo_sapiens/Gene/Variation_Gene/Table?db=core;g=ENSG00000112164;r=6:39048781-39091303;t=ENST00000373256

Protein Data Bank (GLP1R Structure)
PDB ID: 5NX2
https://www.rcsb.org/structure/5NX2

RCSB 3D Structure Viewer
https://www.rcsb.org/3d-view/5NX2/1

mCSM-lig Protein Ligand Affinity Change
https://biosig.lab.uq.edu.au/mcsm_lig/prediction

Video Tutorial
https://drive.google.com/file/d/1ysGrUuLy_ZlU7dvcvMvshptzlMXPL8TE/view?usp=sharing

Purpose

The purpose of this guide is to explore whether genetic variation in the GLP-1 receptor may affect how it interacts with GLP-1 medications by identifying ligand-binding residues and analyzing variants at these positions for their effect on ligand-binding.

Background

GLP-1 medications such as semaglutide (Ozempic) have become a more commonly used medication and are in high demand due to their effectiveness in resulting in significant weight loss.[1−2] GLP-1 stands for Glucagon-like peptide 1, which is a natural hormone in the body that is released after eating. It has many metabolic effects such as increasing insulin secretion when blood glucose levels are high, delaying gastric emptying and reducing food intake.[1−2]

Semaglutide falls into a class of drugs called GLP-1 receptor agonists which work by mimicking the GLP-1 hormone. As it is a peptide-based agonist, it interacts with the GLP-1 receptors (GLP1R) in a similar way to the hormone and can activate the receptors to produce those same effects.[1−2,4] As a result, studies have shown that people can lose around 15% to 25% of their body weight while using these drugs which can be compared to weight loss seen with procedures like gastric sleeve surgery.[2−3]

However, these medications do not work the same for everyone. While many people have benefited from their use, others have also reported that it does not work as effectively, with some even experiencing weight gain.[3] Differences in response between individuals suggests that factors such as genetic variation in the GLP1R may influence how effectively these drugs bind and function. To study this, a bioinformatic approach is necessary to analyze both the structure of the GLP1R, including its interaction with a bound peptide agonist, and genetic variation within the GLP1R gene.

Manual overview

Throughout the guide, the primary tools used for this analysis was the RCSB Protein Data Bank, Ensembl Genome browser, and mCSM-lig. These tools combined allow the connection between protein structure (how the agonist binds to receptor) and genetic variation (how that binding might be affected).

RCSB Protein Data Bank (PDB)

To begin, the RCSB Protein Data Bank (PDB) provides 3D structures of proteins and was used to find a 3D structure of the GLP-1 receptor in order to examine how it interacts with GLP-1 medications. The structure chosen was PDB ID: 5NX2 (https://www.rcsb.org/structure/5NX2) because it includes a bound peptide agonist.[1−2,4] Since GLP-1 medications are also known as agonists, it makes this structure serves as a relevant model for studying these interactions.

The 3D structure viewer feature also in PDB was used to explore the receptor and its components visually, including the polymer (receptor), the ligand (agonist), and carbohydrates. To focus on the interactions between the ligand and surrounding receptor residues, the "components" section in the 3D viewer was used to hide non-relevant components. The ligand-binding sites were then identified by the dashed lines shown in the viewer, which indicate contact between the agonist and the receptor residues. These residues correspond to the amino acids involved in ligand binding**.** Overall, the PDB structure viewer was used to identify a total of five ligand-binding positions within the GLP-1 receptor.

Ensembl Genome Browser / Variant Table

Once the binding positions were identified, they were recorded and the positions were examined in Ensembl to determine whether genetic variation occurs at those sites. So, for the second part of the guide Ensembl was used as a genome browser to examine the human GLP1R gene (https://www.ensembl.org/Homo_sapiens/Gene/Summary?db=core;g=ENSG00000112164;r=6:39048781-39091303;t=ENT00000373256) and access genetic variation data. After searching for the gene (Gene ID: ENSG00000112164), Ensembl has a genetic variation section, and it includes a variant table within it with data gathered from databases such as dbSNP and other genomic resources. This table includes information such as variant type, amino acid coordinates, amino acid changes, and predictions of variant impact, including SIFT and PolyPhen scores. Each row in the table represents a variant and there can be multiple variants at the same position shown as an additional row.

Therefore, the ligand-binding positions identified from the PDB structure were compared to the "AA coord" (amino acid coordinate) and "AA" (amino acid change) column of the variant table which a match would indicate the presence of a variant at that position. So, variants were identified at four of the five ligand-binding positions but there was a total of five variants observed (one position had two variants). These variants were then evaluated using SIFT and PolyPhen scores from the "SIFT" and "PolyPhen" column of the table to determine which were predicted to be damaging.

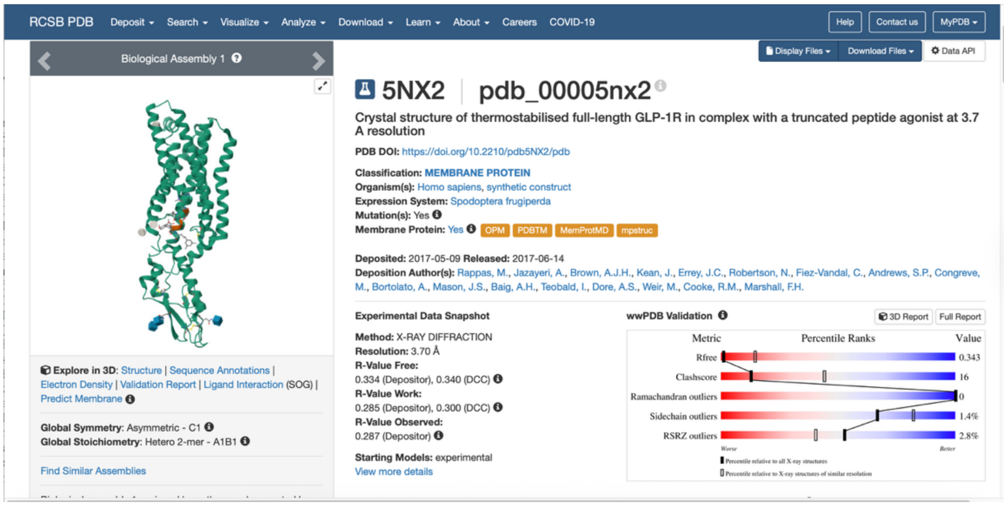
mCSM-ling Protein Ligand Affinity Change

For the final part of the guide, only those variants that were predicted to be damaging were selected for further analysis using mCSM-lig (https://biosig.lab.uq.edu.au/mcsm_lig/prediction). It is a tool used to predict how amino acid changes caused by genetic variants may affect receptor–ligand interactions by calculating changes in binding affinity.[6] The same GLP1R structure (PDB ID: 5NX2) was used for this analysis and uploaded into the tool. Each selected variant was entered separately with the mutation information (Wild-type AA + position + mutant AA), the mutation chain (Chain A), the ligand in the structure (SOG), and a positive value was entered for wild-type affinity to meet input requirements. It then output a predicted affinity change for the variants which helped show how each variant may affect ligand binding.

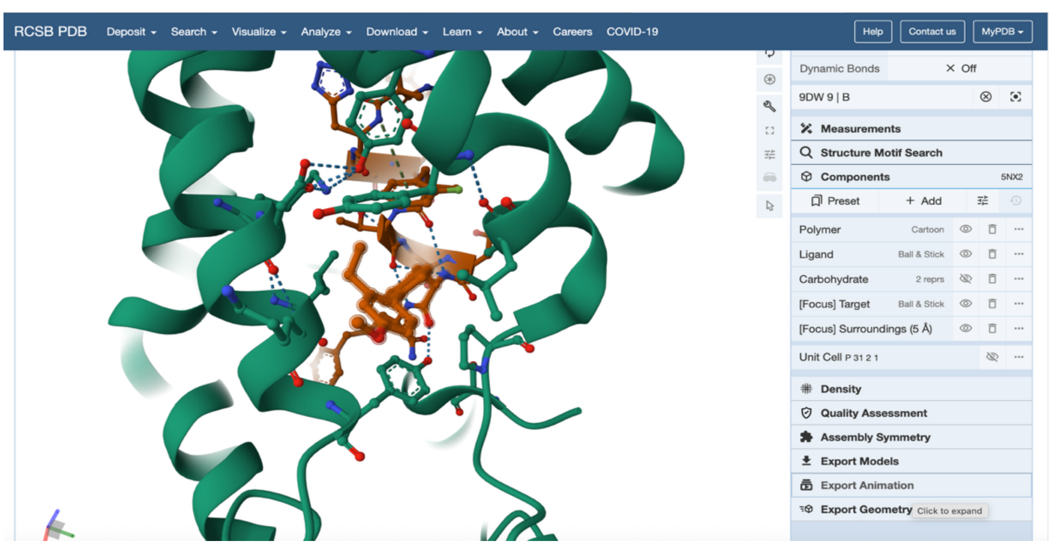
Manual

Part 1: Identify ligand-binding residues using PDB

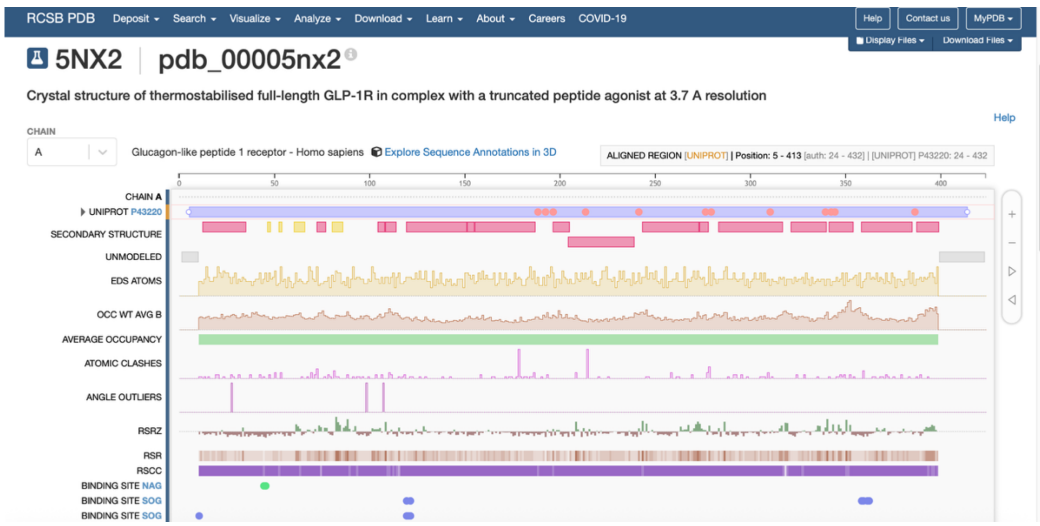
1. Search for GLP1R in RCSB Protein Data Bank (PDB)
    - Go to the RCSB PDB website
    - Enter gene: **GLP1R** in the search bar
    - The search will result in multiple GLP1R structures
2. Choose a structure with a bound **agonist**
    - To visualize the receptor-agonist interaction at the binding site
    - The specific GLP1R structure used in this analysis was **PDB ID: 5NX2** (https://www.rcsb.org/structure/5NX2)



3. Open the 3D structure viewer for 5NX2
   - Select "Explore in 3D Structure"
   - Identify the different components in structure:
     - Polymer (receptor): green
     - Ligand (agonist): orange
     - Carbohydrate: blue
4. Filter components to focus on the interaction
   - In the **Components** section, adjust the visibility of elements in the structure
   - Make the ligand and nearby residues visible
     - Use [Focus] Surroundings (5 Å) to show residues close to ligand
   - Hide non-relevant components such as carbohydrate to reduce visual clutter



5. Locate residues in contact with the ligand
    - Use dashed lines or direct contact to identify interactions
    - Record the names and positions of these residues
6. Organize identified residues and compare numbering
    - Arrange the residues in table
    - Compare the **PDB numbering and "auth" numbering** for each residue



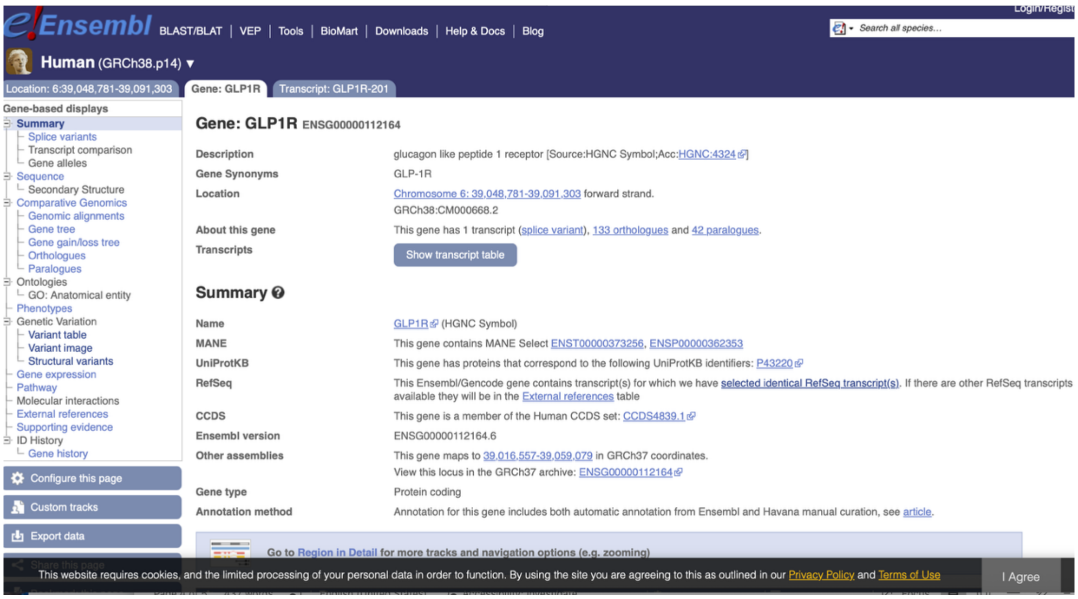
        - These values may differ because PDB does not use the same numbering as the full protein in Ensembl. For example, the sequence alignment shows that PDB positions 5-413 correspond to UniProt/Ensembl positions 24-432 which shows that the numbering is shifted

Residue Name	PDB Position	Auth Position
TRP	14	33
TYR	133	152
ARG	171	190
ASN	281	300
ASP	353	372

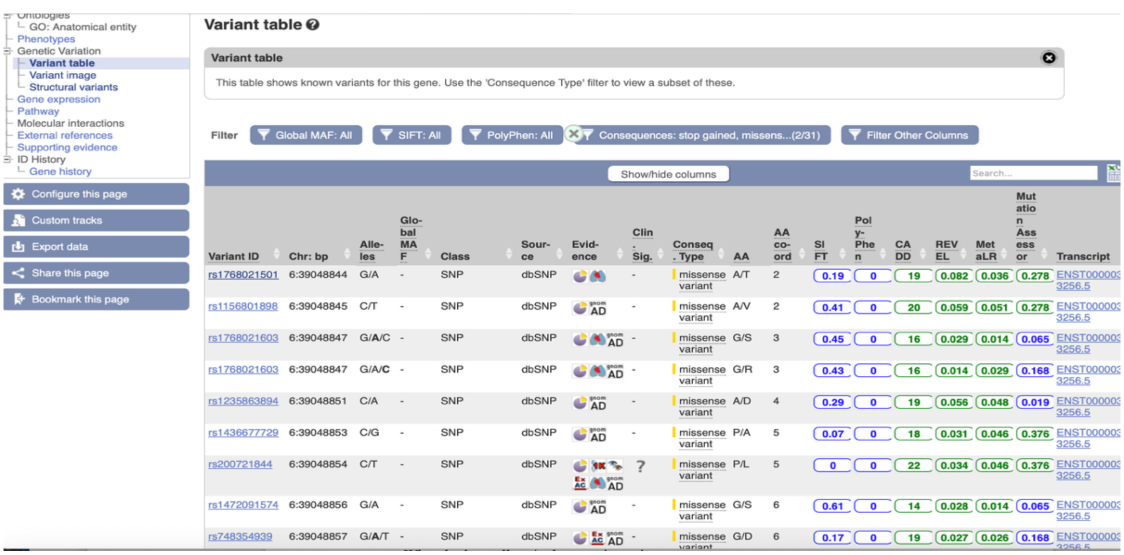
Part 2: Determine whether variants exist at identified binding positions using Ensembl

7. Access the GLP1R gene in Ensembl
    - Go to the Ensembl Genome Browser

- Enter GLP1R in the search bar
    - Select the **human GLP1R gene** (ENSG00000112164): https://www.ensembl.org/Homo_sapiens/Gene/Summary?db=core;g=ENSG00000112164;r=6:39048781-39091303;t=ENST00000373256



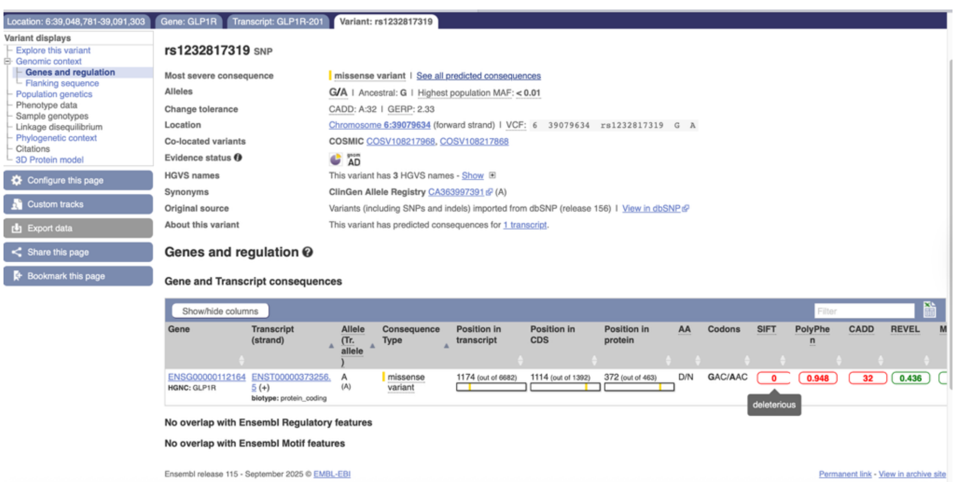
8. Navigate variant data for GLP1R
    - From the GLP1R gene page, select the "Genetic Variation" section
    - Open the **variant table**
        - The variant table involves known genetic variants in the GLP1R gene, including their positions, amino acid changes, and predicted effects

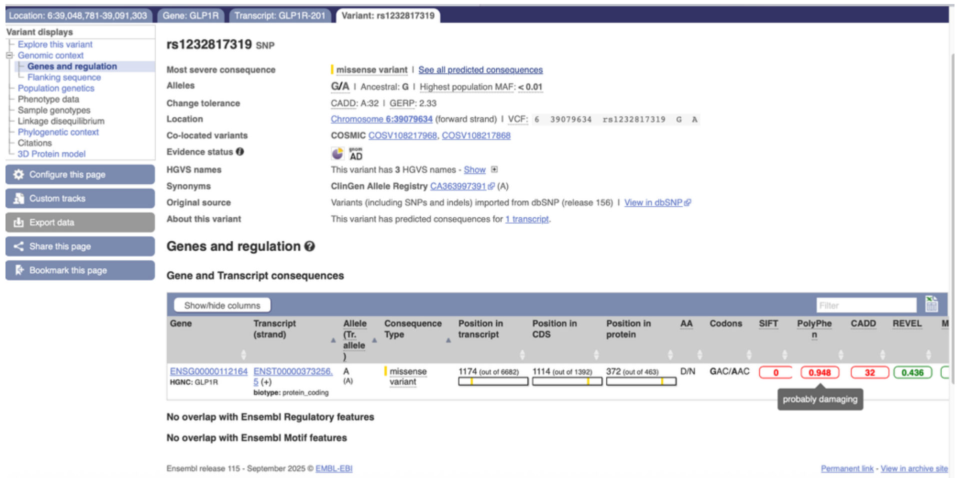


9. Locate variants at identified variant residue positions
    - Use the **AA coord** column to identify variant positions
    - Compare these to the 3D structure positions (auth numbering)
        - Identify variants at matching positions
        - Record amino acid changes and consequence type

Residue Name	Auth Position	Variant Present	AA Change	Consequence Type
TRP	33	Yes	W/R	Missense variant
TYR	152	Yes	Y/*	Stop gained
ARG	190	Yes	R/Q	Missense variant
ARG	190	Yes	R/*	Stop gained
ASN	300	No	----------	----------
ASP	372	Yes	D/N	Missense variant

10. Analyze predicted variant impact using **SIFT and PolyPhen**
    - Locate the SIFT and PolyPhen scores for *only* identified variants from step 9
        - Also found in the variant table on Ensembl
        - Record SIFT and PolyPhen values for these variants
            - Select the variant ID column in table to access additional information for each variant
            - Example:





- Use SIFT and PolyPhen predictions to determine whether the identified amino acid changes are likely to be damaging or not
  - SIFT[5]: lower score = more likely damaging
    - Less than 0.05 = deleterious
    - Greater than or equal to 0.5 = tolerated
  - PolyPhen[5]: higher score = more likely damaging
    - Greater than 0.9 = probably damaging
    - Greater than 0.5 but less than 0.9 = possibly damaging
    - Less than or equal to 0.5 = benign

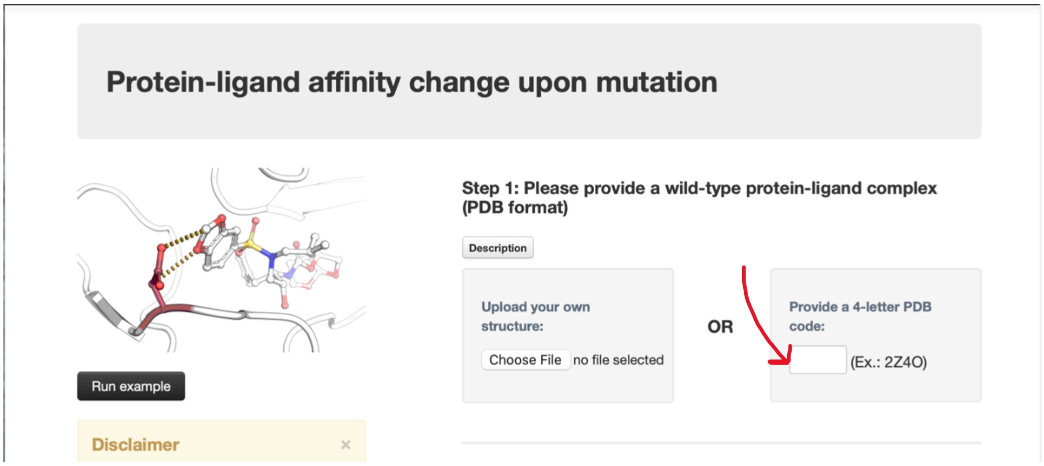
Auth Position	AA Change	SIFT	PolyPhen	Interpretation
33	W/R	0.19	0.017	Tolerated / benign
152	Y/*	----------	----------	Cannot be analyzed
190	R/Q	0	0.997	Predicted damaging
372	D/N	0	0.948	Predicted damaging

Part 3: Predict the impact of damaging variants on ligand binding using mCSM-lig

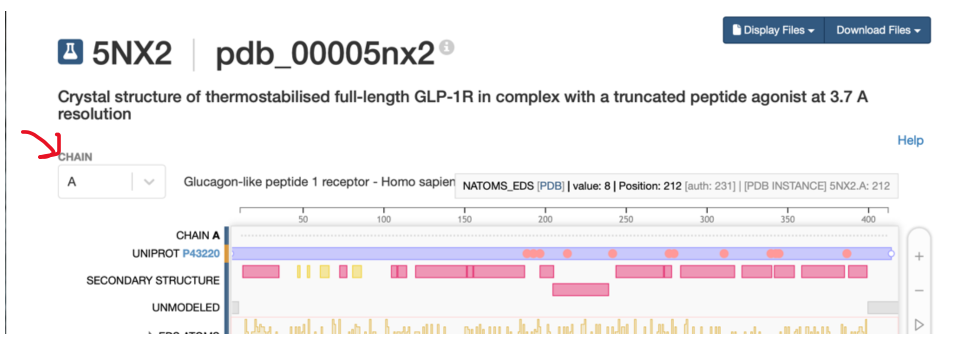
11. Upload the GLP1R PDB structure

- Open mCSM.lig (https://biosig.lab.uq.edu.au/mcsm_lig/prediction)

- In the step 1 section, write **5NX2** (the PDB ID for the GLP1R structure)



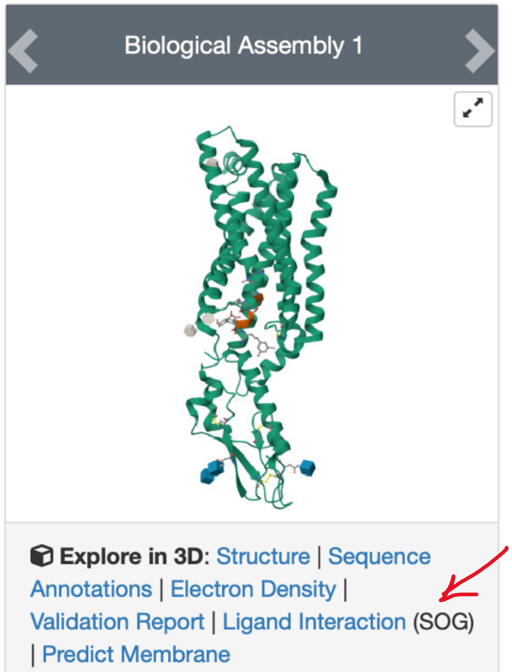
12. Input variant information

- In the step 2 section, for each selected variant enter the mutation and mutation chain at separate times
    - Variant at position 190
        - Mutation: **R190Q** (using auth position and AA change from previous table)
        - Mutation chain: **A** (in PDB the GLP1R is Chain A)
        - 

    - Variant at position 372
        - Mutation: **D372N**
        - Mutation chain: **A**

13. Input ligand information

- Find ligand ID from PDB structure
    - In the PDB structure 5NX2, the ligand is labeled as **SOG**



14. Enter wild-type affinity

- Enter a **small positive value** (greater than 0) to satisfy input requirement
    - For example: wild-type affinity = 1

15. Submit prediction for each variant and record output

- Variant at position 190 output:

Predicted Affinity Change:
-1.057 log(affinity fold change) - *Destabilizing*

Mutation information:
Wild-type: **R**
Position: **190**
Mutant-type: **Q**
Chain: **A**
Ligand ID: **SOG**
Distance to ligand: **18.931 Å**
DUET stability change: -1.704 Kcal/mol

  - Negative change in binding affinity = decreased binding affinity

- Variant at position 372 output:

Predicted Affinity Change:
-1.175 log(affinity fold change) - *Destabilizing*

Mutation information:
Wild-type: **D**
Position: **372**
Mutant-type: **N**
Chain: **A**
Ligand ID: **SOG**
Distance to ligand: **8.854 Å**
DUET stability change: -0.787 Kcal/mol

  - Negative change in binding affinity = decreased binding affinity

Conclusion

The results show that genetic variation does occur at different ligand-binding positions in the GLP-1 receptor, including positions 33, 152, 190, and 372. Most of the variants are missense, while one is a stop-gained mutation. The SIFT and PolyPhen predictions suggest that some of these variants may be damaging, especially at positions 190 and 372. Further analysis through mCSM-ling predicted that these variants at positions 190 and 372 result in negative changes in binding affinity. This can suggest that these variants may decrease ligand binding which in this analysis could mean that the GLP-1 medications might not bind or interact as well with the GLP-

1 receptor, leading to potential decrease in effectiveness of the drugs. However, these results are based on predictions as the tools used have limitations because they do not directly show actual structural changes. Overall though, this analysis could suggest that genetic variation in the GLP-1 receptor could be able to influence how effectively GLP-1 medications bind and function, which may help explain why different individuals respond differently to these drugs. It also suggests that genetic variation is an important factor to consider when studying drug effectiveness for other medications as well.

References

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6. Pires, Douglas E V et al. “mCSM: predicting the effects of mutations in proteins using graph-based signatures.” *Bioinformatics (Oxford, England)* vol. 30,3 (2014): 335-42. doi:10.1093/bioinformatics/btt691